Data Storytelling

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Week 06

Announcements

- Adjusted syllabus to reflect current progress
- How are the assignments coming along? Due tonight at 11:59 PM!
- How do you generate a pdf report from a qmd file?
  - Specify output format in the YAML header
    - Eg. `format: pdf`
  - Click the "render" button in RStudio
  - Note: If you are using R Markdown, you'll use `output: pdf` instead.

Predicting Data

Why Predict Data?

- One of the most valued skills data analysts is how to predict outcomes.
  - What will the stock market do tomorrow?
  - What will the weather be like next week?
  - Predicting job growth in the following quarter.
- These predictions are valuable for decision-making, planning, and understanding trends.

How can we predict data?

- Typically, we use statistical models, machine learning algorithms, or time series analysis to make predictions based on historical data.
  - Models are some function mapping input data to output data.

$$Y = f(X) + \epsilon$$

- Where,
  - Y is the outcome we want to predict (ex. GDP, stock price, etc.),
  - X is the input data (ex. historical GDP, stock prices, etc.),
  - f is the function that maps X to Y,
  - ϵ is the error term (unexplained variance)

Modeling Approaches

- The model you use depends on the type of data you have and the problem you're trying to solve.
  - **Continuous outcomes**:
    - *Examples:* GDP, temperature, company revenue, etc.
    - *Models:* Linear regression, support vector regression, or transformer models.
  - **Binary outcomes**:
    - *Examples*: Will a war occur between two countries? Will a person vote for a particular candidate? Will a company go bankrupt?
    - *Models*: Logistic regression, decision trees, or random forests.
  - **Multiclass outcomes**:
    - *Examples*: What type of job will a person have? What type of product will a customer buy? What type of disease does a patient have?
    - *Models*: Multinomial logistic regression, transformer models.

Linear Regression

- Linear regression, or Ordinary Least Squares (OLS), is a simple and widely used method for predicting a continuous outcome variable based on one or more predictor variables.
- Goal is to fit a line to observed data. Does anyone remember the formula for a line?

$$Y = mX + b$$

- In linear regression, we can rewrite this as:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

A fitted Linear Model

- Let's say that we wanted to model the relationship between study time (X) and test scores (Y). We could fit a linear regression model to our data, which would give us estimates for β_0 and β_1.

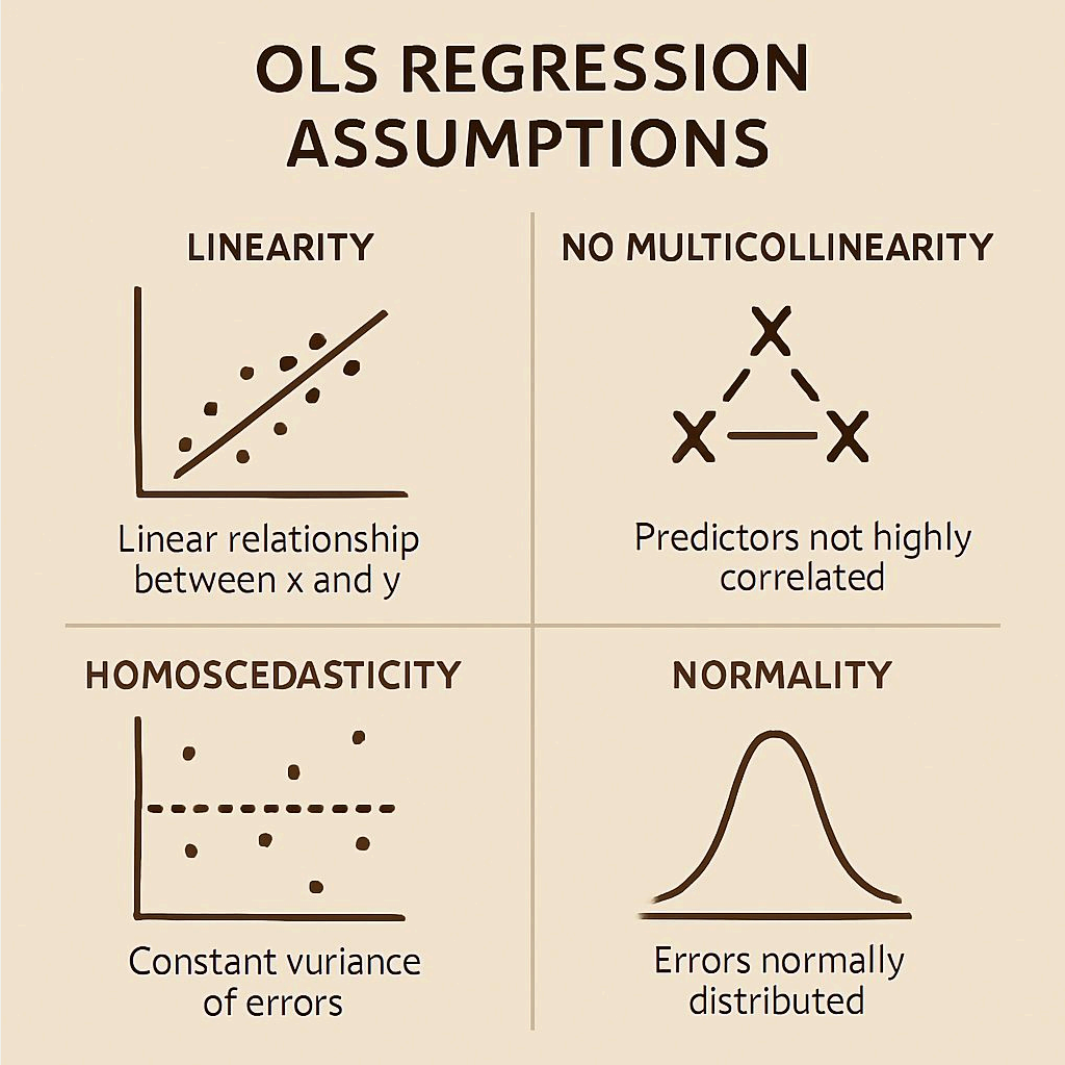
$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$$

- How would we interpret $\hat{\beta}_0$ and $\hat{\beta}_1$ in this context?

Note: Hypothesis Testing vs. Prediction

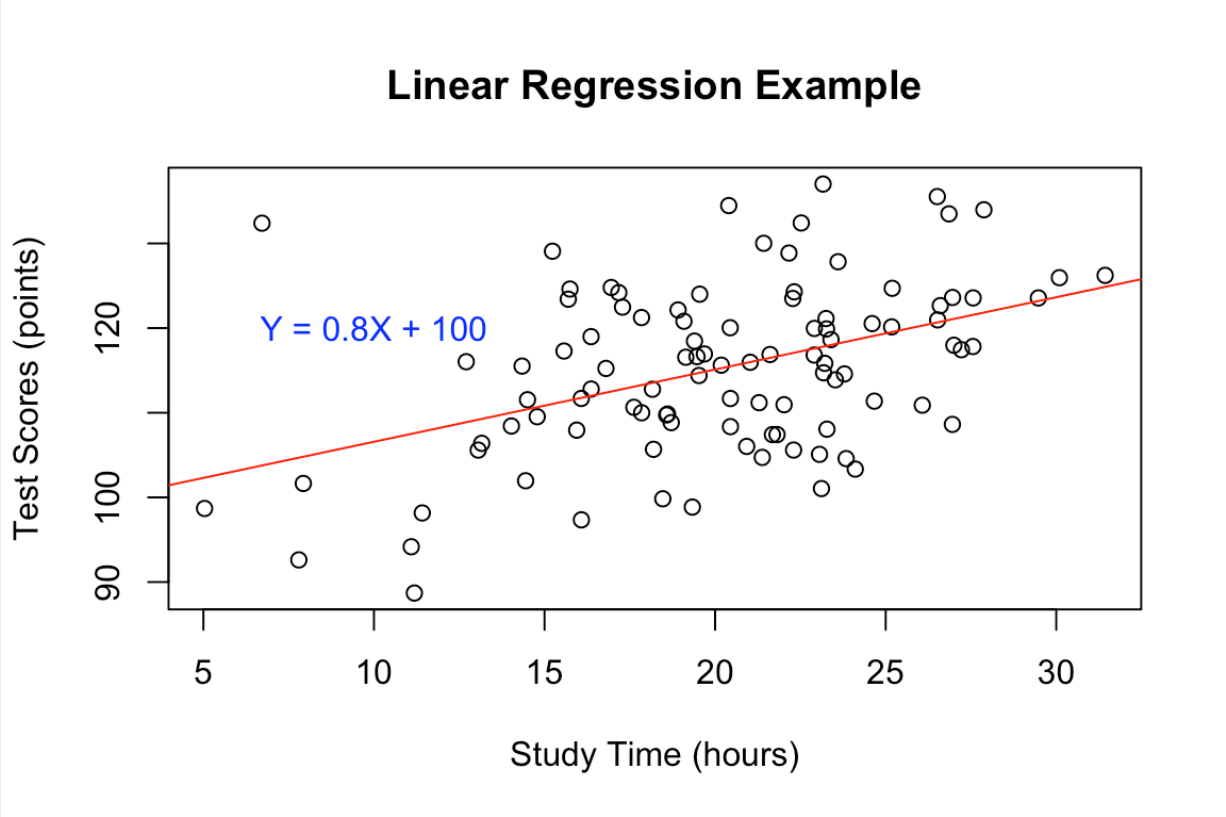
- *Hypothesis Testing:* Focuses on understanding the relationship between variables and testing specific hypotheses about the coefficients (e.g., is β_1 significantly different from zero?).
  - Interpretation of p-value:
    - A low p-value (typically < 0.05) indicates that we can reject the null hypothesis, suggesting that there is a statistically significant relationship between the predictor and outcome variable.
    - A high p-value suggests that we fail to reject the null hypothesis, indicating that there is not enough evidence to support a significant relationship.
- *Prediction:* Focuses on accurately predicting the outcome variable for new data, often prioritizing model performance metrics over statistical significance of coefficients.

Linear Regression Assumptions



Linear Regression Illustration

```r
set.seed(42)
X <- rnorm(100, mean=20, sd=5) # Study time in hours
Y <- 0.8 * X + rnorm(100, mean=100, sd=10)
plot(X, Y, main="Linear Regression Example", xlab="Study Time (hours)", ylab="Test Scores (points)")
text(10, 120, "Y = 0.8X + 100", col="blue")
abline(lm(Y ~ X), col="red")
```



Linear Regression Illustration

```r
summary(lm(Y ~ X))
```

```
Call:
lm(formula = Y ~ X)

Residuals:
     Min       1Q   Median       3Q      Max 
-18.8842  -5.0664   0.1225   5.4106  28.6240 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  98.0300     3.6499  26.858  < 2e-16 ***
X             0.8543     0.1753   4.873 4.24e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.083 on 98 degrees of freedom
Multiple R-squared:  0.195, Adjusted R-squared:  0.1868 
F-statistic: 23.74 on 1 and 98 DF,  p-value: 4.238e-06
```

Evaluation of Linear Regression

- Given that you want to predict a continuous variable, how do you evaluate how good of a job your model is doing?
  - *R-squared:* Proportion of variance in the outcome variable that is explained by the predictor variable(s).
  - *Mean Absolute Error (MAE):* Average absolute difference between predicted and actual values.
  - *Mean Squared Error (MSE):* Average squared difference between predicted and actual values.

Training and Testing Data

- To evaluate the performance of our model, we typically split our data into a training set and a testing set.
  - The training set is used to fit the model, while the testing set is used to evaluate its performance on unseen data.
- A common split is 80% for training and 20% for testing, but this can vary depending on the size of the dataset and the specific problem you're trying to solve.
  - Your metrics are applied to the testing set to evaluate how well your model generalizes to new data.
  - Failure to properly split data can lead to overfitting
    - Where the model performs well on the training data but poorly on new, unseen data.

Training and Testing Data Illustration

```r
set.seed(42)
X <- rnorm(1000, mean=20, sd=5) # Study time in hours
Y <- 0.8 * X + rnorm(1000, mean=100, sd=10)
training_indices <- sample(1:1000, size=800)
X_train <- X[training_indices] # Training data for predictor variable
Y_train <- Y[training_indices] # Training data for outcome variable
X_test <- X[-training_indices] # Testing data for predictor variable
Y_test <- Y[-training_indices] # Testing data for outcome variable
model <- lm(Y_train ~ X_train) # Fit the linear regression model using the training data
summary(model)
```

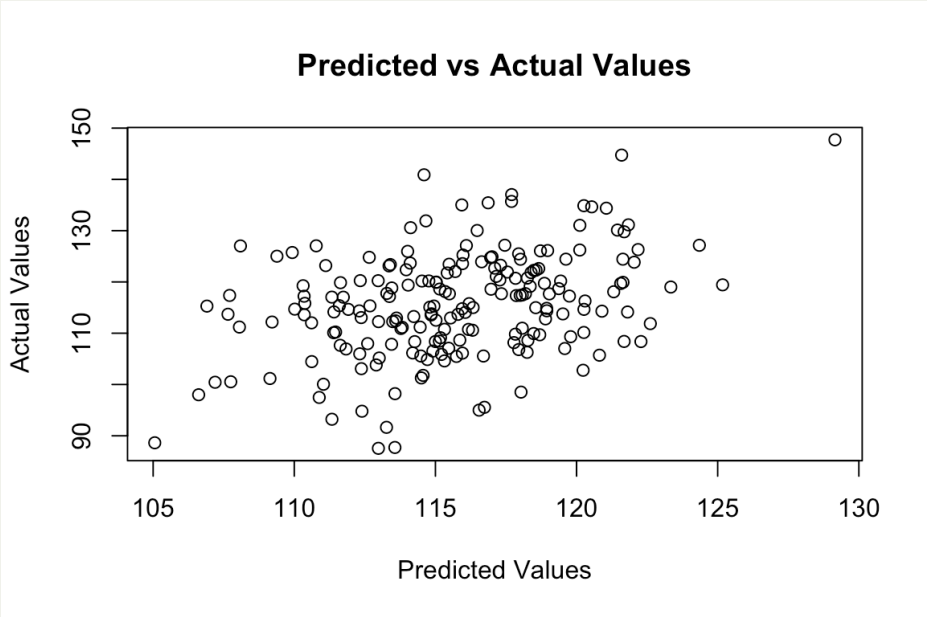
```
Call:
lm(formula = Y_train ~ X_train)

Residuals:
    Min      1Q  Median      3Q     Max 
-29.242  -6.579  -0.078   6.716  35.857 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 99.54027    1.42859   69.68   <2e-16 ***
X_train      0.82133    0.06952   11.81   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.965 on 798 degrees of freedom
Multiple R-squared:  0.1489,	Adjusted R-squared:  0.1478 
F-statistic: 139.6 on 1 and 798 DF,  p-value: < 2.2e-16
```

Evaluating the Model

```r
predictions <- predict(model, newdata=data.frame(X_train=X_test)) # Generate predictions on the testing data
r_squared <- cor(predictions, Y_test)^2 # Calculate the correlation between predictions and actual values in the testi
print(paste("R-squared:", round(r_squared, 2)))
```

```
[1] "R-squared: 0.14"
```

```r
plot(predictions, Y_test, main="Predicted vs Actual Values", xlab="Predicted Values", ylab="Actual Values")
```



- How does this compare with the training data? Is the model performing well on the testing data?

Logistic Regression

- Logistic regression is a statistical method for predicting a binary outcome variable based on one or more predictor variables.
  - Works similarly to linear regression, but instead of fitting a line, it fits an S-shaped curve (sigmoid function) to the data.
- The formula for logistic regression is:

$$P(Y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

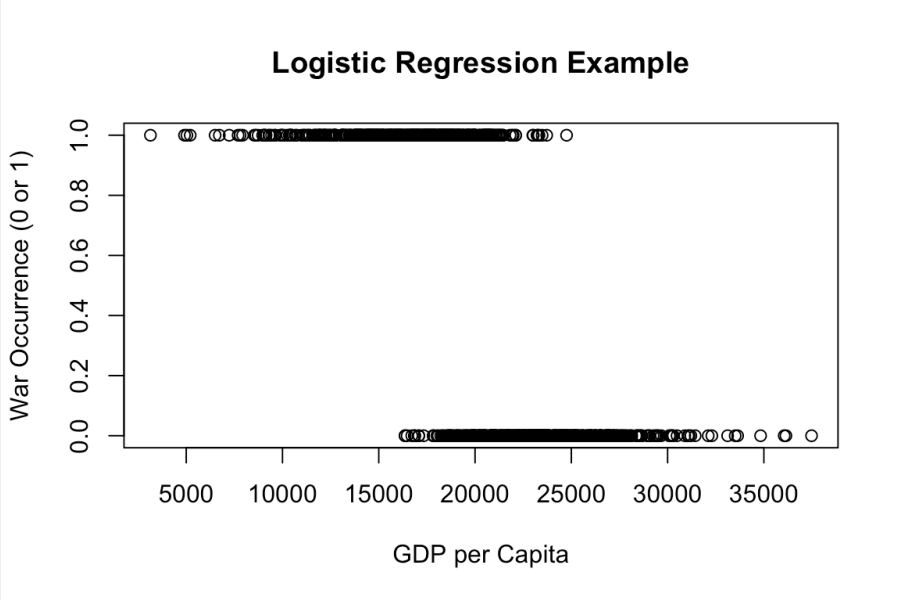
- Where,
  - $P(Y = 1)$ is the probability of the outcome variable being 1 (ex. war occurring, person voting for a candidate, etc.),
  - β_0 is the intercept,
  - β_1 is the coefficient for the predictor variable X.

Evaluating Logistic Regression

- To evaluate the performance of a logistic regression model, we can use metrics such as:
  - *Accuracy:* Proportion of correct predictions (both true positives and true negatives) out of all predictions.
  - *Precision:* Proportion of true positives out of all positive predictions.
  - *Recall (Sensitivity):* Proportion of true positives out of all actual positives.
  - *F1 Score:* Harmonic mean of precision and recall, providing a balance between the two.

Example: Predicting War with Logistic Regression

```r
set.seed(42)
#GDPPC
GDPPC <- rnorm(1000, mean=20000, sd=5000) # GDP per capita
#War occurrence (binary outcome)
War <- 1/(1 + exp(-(10 + -0.0005 * GDPPC + rnorm(1000, mean=0, sd=1)))) # Probability of war occurrence based on GDP pe
War <- ifelse(War > 0.5, 1, 0) # Convert probabilities to binary outcomes (1 for war, 0 for no war)
data <- data.frame(GDPPC, War)
plot(data$GDPPC, data$War, main="Logistic Regression Example", xlab="GDP per Capita", ylab="War Occurrence (0 or 1)")
```



Fit the logistic regression model

```r
train_indices <- sample(1:1000, size=800)
train_data <- data[train_indices, ] # Training dataset
test_data <- data[-train_indices, ] # Testing dataset
model <- glm(War ~ GDPPC, data=train_data, family="binomial") # Fit logistic regre
summary(model)
```

```
Call:
glm(formula = War ~ GDPPC, family = "binomial", data = train_data)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.704e+01  1.303e+00   13.08   <2e-16 ***
GDPPC       -8.480e-04  6.466e-05  -13.12   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1108.54  on 799  degrees of freedom
```

Example: Evaluating Logistic Regression Model

```r
predictions <- predict(model, newdata=test_data, type="response") # Generate predicted probabilities on the testing dat
predicted_classes <- ifelse(predictions > 0.5, 1, 0) # Convert probabilities to binary predictions
accuracy <- mean(predicted_classes == test_data$War) #
table(predicted_classes, test_data$War) # Confusion matrix
```

```
predicted_classes  0  1
                0 81  9
                1 20 90
```

```r
print(paste("Accuracy:", round(accuracy, 2)))
```

```
[1] "Accuracy: 0.86"
```

Example: Predicting GDP with Light Data

- *Why would we want to predict GDP?*
  - Some countries produce GDP data that is unreliable – North Korea, for example.
  - Data reporting may be interrupted by war, natural disasters, or political instability.
- *What kind of variable is GDP?*
  - A continuous variable
- *What kind of model would we use to predict GDP?*
  - Linear regression, support vector regression, or transformer models.

Pitfalls of Data Analysis

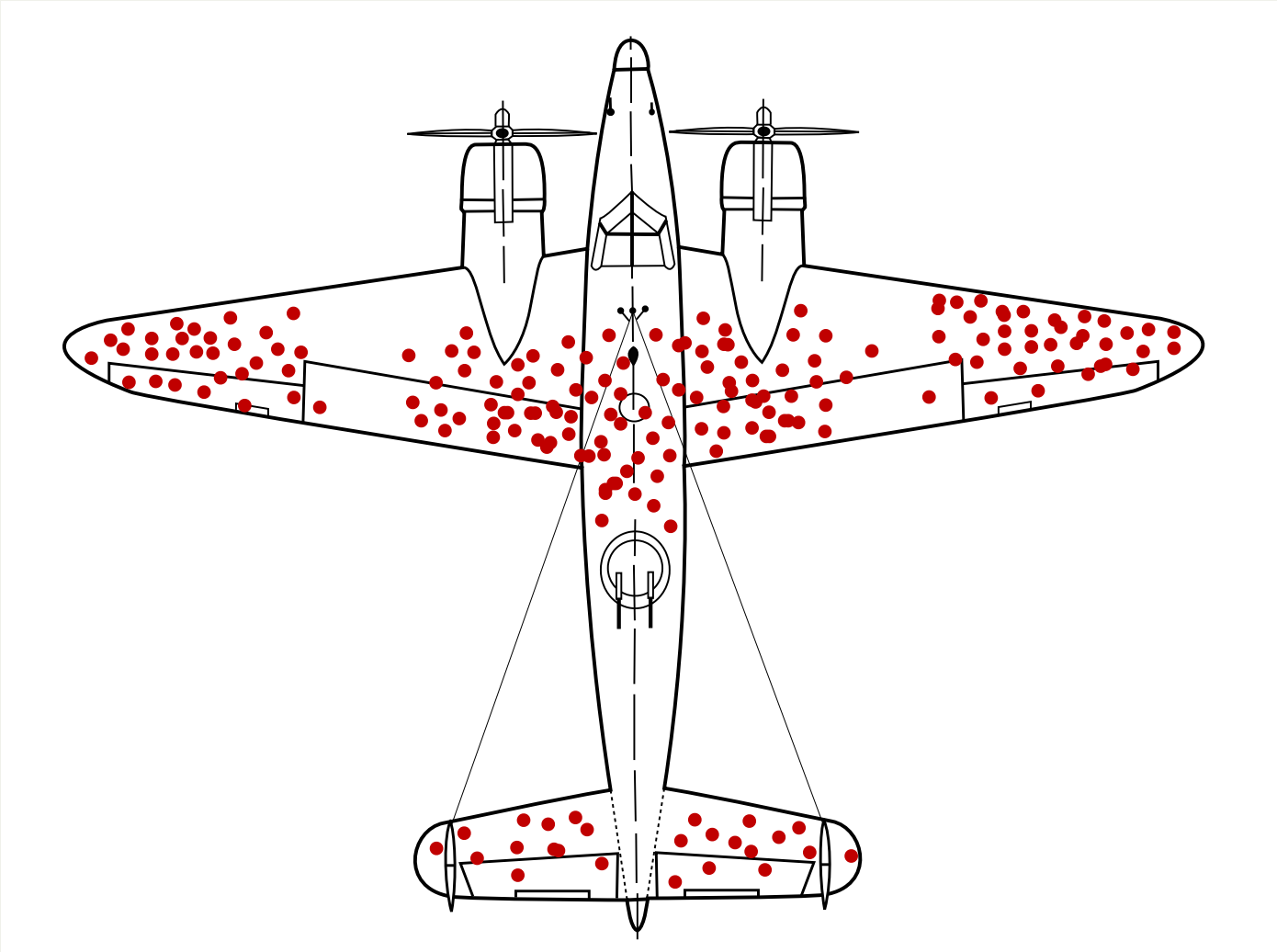
Outliers

- Outliers are data points that are significantly different from the rest of the data. They can occur due to measurement errors, data entry errors, or they may be genuine observations that are rare or extreme.
  - If the outlier is due to a data entry or measurement error, it should be removed from the dataset to prevent it from skewing the analysis.
  - If the outlier is a byproduct of the data production process, it should be retained in the dataset, as it may provide valuable information about the underlying phenomenon being studied.

Survivorship Bias

- Survivorship bias occurs when we focus on the successful cases or “survivors” of a process while ignoring the unsuccessful cases or “non-survivors.” This can lead to a distorted view of reality and incorrect conclusions.
  - For example, if we only study successful companies, we may conclude that certain business strategies are effective, while ignoring the many companies that failed using those same strategies.

Plane Example of Survivorship Bias



Exercise time!

- Next, let's apply what we've learned about predicting data and evaluating models to a real-world dataset.